

Types of Modulators:

Direct - External modulations:

Direct modulation switch the laser intensity by changing the driving current to the laser.

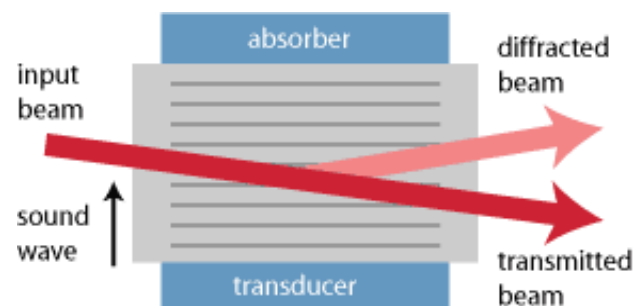
Good for simple and cheap applications, but not suitable for high speed and long distance modulation due to switching effect of semiconductor technology, and the chirping effect (chirping effect is the instability in signal frequency due to variation of laser intensity).

External modulation is caused by supplying the light out of the laser source into modulators, and then into the channel, modulators alter the properties of the light signal.

Good for high speed and less distortion applications, but it's very complicated method, more expensive than direct modulation, multiple variables must be considered before using this method (like wavelength variation, temperature variation, industry fabrication imperfections, etc).

Types of External Modulators:

- **Electro-Optic Modulators:** they are a type of external modulator that uses electric field to modify the refractive index of waveguide, in which it changes the properties of the optical signal passing through the wave guide, the properties in effect are amplitude, phase, frequency, polarization or a combination of the previous elements.
 1. The phase change happens when the **electric field simply changes the refractive index of the path**, which changes the path length and thus phase.
 2. **The amplitude change happens** when the electric field modifies the material characteristic and make it absorb part of the light signal or by interference of mach-zehnder implementation.
 3. Mach-zehnder interferometer make changes through the light signal by splitting the signal and making phase shift in each part of the signal, then recombining the two parts for constructive or destructive interference.
 4. Polarization modulation happens by controlling polarization state of light.
- **Acousto-Optic Modulators:** it is a type of external modulators that uses bragg diffraction principle to control properties of light passing through certain material using acoustical waves, the properties that can be modified are amplitude, phase, frequency and beam deflection.



1. The amplitude change happens by varying the amplitude of the acoustic wave, which changes the diffraction efficiency.
 2. The frequency modulation happens by adjusting the frequency of the acoustical signal, which adds or subtracts to the value of the optical signal frequency.
 3. The phase modulation happens by varying the phase of the acoustical signal, which affects the diffracted beam phase.
 4. The beam deflection happens by varying the frequency of the acoustical signal, in which it changes the bragg angle of diffraction.
- Electro-Absorption Modulators: a type of external modulators that controls the amplitude and wavelength spectrum of the optical signal by controlling the absorption rate of the waveguide material using electrical field.
 1. The amplitude modulation is occurred by controlling how much light is absorbed in the waveguide using electrical field.
 2. Phase modulation occurs uncontrolled due to the effect of electrical field in controlling the phase of the optical signal due to changing of the refractive index of the waveguide.
 3. The wavelength modulation happens by tuning the voltage applying electrical field, in which the absorbed light wavelength changes, thus the remaining optical signal wavelength spectrum changes.
 - Magneto-Optic Modulators: type of external modulators that controls the amplitude and polarization using magnetic fields by the principle of faraday effect.
 1. Amplitude modulation occurs when combining a polarizer and modulator, by rotating the polarizer, the signal varies and thus the amplitude varies.
 2. Polarization modulation occurs by altering the polarization of a light using the magnetic field.
 - Thermo-Optic Modulators: it is a type of external modulators that controls refractive index of waveguide using heat energy.
 1. Phase modulation is achieved by simply heating the waveguide to change its refractive index and thus the signal phase.
 2. Amplitude modulation happens in MZI by phase shifting the signals which causes constructive or destructive interference.
 3. Wavelength modulation occurs when changing the phase of the signal which changes the resonant wavelength in ring resonators and bragg gratings.

Type	Modulation Method	Common Material	Application
Electro-Optic (EOM)	Refractive index (electric field)	LiNbO ₃ , GaAs	High-speed telecom
Electro-Absorption	Light absorption	InGaAs, InP	Compact modulators in fiber optics
Acousto-Optic (AOM)	Acoustic wave diffraction	Quartz, TeO ₂	Laser scanning, spectroscopy
Magneto-Optic	Polarization (magnetic field)	Garnets	Optical isolators, sensors
Thermo-Optic	Heat-induced index change	Silicon, Polymers	Integrated photonics

Silicon Photonics Modulators:

- Mach-Zehnder Interferometer Modulators
- Micro Ring Resonator Modulators
- Electro-Absorption Modulators
- Carrier Injection/Depletion based Modulators
- Thermo-Optic Modulators

Modulators:

Micro ring modulators:

Low insertion loss / energy efficient / low size thus they suit for high density fabrications

Fabrication sensitive / temperature variation sensitive (no more than 0.2° C) thus they require heaters and controlling circuits

Mach-Zehnder modulators:

Can sustain temperature changes up to 8.3° C /

Large in size compared to the micro ring modulator

Types: 2X2 1X2 1X1 / symmetric asymmetric (optical path length = arm length x refractive index) / balanced unbalanced (arm length)

References:

1-Types of modulators

Direct vs external modulation

<https://ieeexplore.ieee.org/abstract/document/7158985>

Electro-optic modulation

<https://pubs.aip.org/aip/jap/article/130/1/010901/158529>

Acousto-optic modulation

<https://bibliotekanauki.pl/articles/1377937.pdf>

[Acousto-optical modulation of thin film lithium niobate waveguide devices](#)

Electro-absorption modulation

<https://patents.google.com/patent/US6633425B2/en>

Magneto-optic modulation

[https://www.researchgate.net/profile/Mohsen-Basavad/post/What about TMOKE measuring in transmission configuration not reflection/attachment/5cf4ffb73843b0b982557d98/AS%3A765664788762627%401559560119807/download/10.5923.j.ijea.20170701.03_2.pdf](https://www.researchgate.net/profile/Mohsen-Basavad/post/What+about+TMOKE+measuring+in+transmission+configuration+not+reflection/attachment/5cf4ffb73843b0b982557d98/AS%3A765664788762627%401559560119807/download/10.5923.j.ijea.20170701.03_2.pdf)

Thermos-optic modulation

[Fast thermo-optical modulators with doped-silicon heaters operating at 2 \$\mu\text{m}\$](#)

2- Silicon photonics Modulators

[Polymer modulators in silicon photonics: review and projections](#)

3- Modulators

[High-Speed Silicon Photonics Modulators | IEEE Journals & Magazine | IEEE Xplore](#)